

Loss Reserve Estimation for Indian MET Lines

Chain Ladder and Bornhuetter-Ferguson Methods

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1 Executive Summary

This report presents loss reserve estimates for three non-life insurance lines material to the Indian reinsurance market: Motor (Third Party and Own Damage), Engineering (Industrial All Risk, Construction All Risk, Erection All Risk), and Treaty (proportional and excess-of-loss reinsurance).

Two actuarial reserving methods are applied and compared:

- **Chain Ladder (Mack, 1993)**: A development pattern extrapolation method that projects historical age-to-age factors forward to estimate ultimate losses.
- **Bornhuetter-Ferguson (BF)**: A credibility blend of an a priori expected loss ratio and actual emergence, providing stability for immature and CAT-contaminated accident years.

Key finding: Chain Ladder and Bornhuetter-Ferguson produce materially similar estimates for Motor, where high volumes yield stable development patterns. For Engineering, significant divergence is observed in accident years 2018-19 and 2019-20, attributable to catastrophe contamination from the Kerala floods (2018) and Cyclone Fani (2019). BF is the preferred method for these years. Treaty results are intermediate, with BF preferred for the two most recent accident years.

All triangles in this report are synthetic, generated from parameters calibrated to IRDAI Annual Report 2022-23 and GIC Re Annual Report 2022-23 disclosures.

2 Indian Non-Life Market Context

2.1 Motor: MACT Judicial Lag

Indian Motor Third Party (TP) claims are adjudicated by Motor Accidents Claims Tribunals (MACT). Judicial delays — arising from case backlogs, appellate proceedings, and structured settlement negotiations — mean that bodily injury claims in India typically develop over 5 to 7 years. This is materially longer than western benchmarks (UK Motor TP typically 70-75% emerged by end of development year 2).

The development pattern used in this analysis reflects this lag:

Table 1: Motor: Cumulative Development Pattern

Development Period	% of Ultimate Emerged
Year 1	45%
Year 2	68%
Year 3	80%
Year 4	88%
Year 5	93%

Year 6	96%
Year 7	98%
Year 8	100%

Only 45% of ultimate Motor losses are emerged by the end of development year 1, compared to approximately 60-65% in European markets. This has material implications for tail factor selection and IBNR estimation for recent accident years.

2.2 Engineering: Catastrophe Contamination

Indian Engineering lines experienced three major catastrophe events in rapid succession:

- **Chennai floods (2015)**: Widespread industrial damage across Tamil Nadu
- **Kerala floods (2018)**: The most severe flooding in Kerala in nearly a century, causing significant IAR and CAR losses
- **Cyclone Fani (2019)**: Category 5 cyclone making landfall in Odisha, with material Engineering losses

These events contaminated accident year development patterns. An insurer or reinsurer applying pure Chain Ladder without adjusting for these years will project CAT-inflated link ratios into future non-CAT years, systematically overstating Engineering reserves.

2.3 Treaty: Cedant Aggregation Effect

A treaty reinsurer's loss triangle aggregates experience across multiple cedants. In years with widespread catastrophe activity — such as 2018 — proportional treaty portfolios absorb CAT losses from cedants simultaneously, elevating the treaty loss ratio in a manner that does not reflect the underlying non-CAT development pattern. GIC Re's treaty premium approximated INR 8,000-9,000 Crores as of 2022-23 (GIC Re Annual Report 2022-23).

3 Data

3.1 Triangle Construction

Loss development triangles were constructed synthetically using parameters calibrated to Indian market benchmarks. The simulation methodology follows a lognormal process model:

For each accident year i and development period j , cumulative losses $C_{i,j}$ are drawn from:

$$C_{i,j} \sim \text{Lognormal}(\mu_{i,j}, \sigma)$$

where $\mu_{i,j} = \log(P_i \times ELR_i \times d_j) - \frac{\sigma^2}{2}$, P_i is earned premium for accident year i , ELR_i is the ultimate loss ratio, d_j is the cumulative development pattern at period j , and σ is set to reflect line-specific process variance.

3.2 Parameter Sources

Table 2: Triangle Simulation Parameters by Line

Line	Base Premium (AY2016, INR Cr)	Premium Growth	Ultimate ELR Range	Process CV	Prin
Motor	8,500	~12% p.a.	76-81%	4%	IRD
Engineering	420	~10% p.a.	68-95%	12%	IRD
Treaty	1,200	~10% p.a.	72-88%	7%	GIC

The Engineering ELR range of 68-95% reflects normal year experience (68-72%) versus CAT-contaminated years (AY2018: 95%, AY2019: 85%), consistent with IRDAI Engineering segment combined ratio data.

3.3 Observed Triangles

Table 3: Motor: Cumulative Incurred Losses (INR Crores)

	Dev1	Dev2	Dev3	Dev4	Dev5	Dev6	Dev7	Dev8
AY2016	2,905	4,546	5,093	5,805	6,158	6,580	6,570	6,510
AY2017	3,464	5,001	6,062	6,325	7,306	6,717	7,203	NA
AY2018	3,853	5,671	6,918	7,348	7,392	8,229	NA	NA
AY2019	4,308	6,260	7,021	8,436	8,180	NA	NA	NA
AY2020	4,716	6,962	8,385	9,234	NA	NA	NA	NA
AY2021	5,302	7,817	9,468	NA	NA	NA	NA	NA
AY2022	5,344	9,201	NA	NA	NA	NA	NA	NA
AY2023	6,398	NA	NA	NA	NA	NA	NA	NA

Table 4: Engineering: Cumulative Incurred Losses (INR Crores)

	Dev1	Dev2	Dev3	Dev4	Dev5	Dev6	Dev7	Dev8
AY2016	136.2	209.8	245.1	243.4	270.3	234.2	255.8	225
AY2017	182.5	263.7	283.0	349.7	282.6	309.3	386.4	NA
AY2018	256.5	475.1	404.8	512.4	554.6	491.0	NA	NA
AY2019	286.2	407.0	461.1	464.5	468.3	NA	NA	NA
AY2020	252.8	351.7	356.7	565.9	NA	NA	NA	NA
AY2021	306.9	394.5	370.1	NA	NA	NA	NA	NA
AY2022	232.5	374.0	NA	NA	NA	NA	NA	NA
AY2023	323.3	NA	NA	NA	NA	NA	NA	NA

Table 5: Treaty: Cumulative Incurred Losses (INR Crores)

	Dev1	Dev2	Dev3	Dev4	Dev5	Dev6	Dev7	Dev8
AY2016	416	633	753	797	795	926	781	819
AY2017	473	722	878	945	986	1,000	904	NA
AY2018	594	967	1,045	1,135	1,270	1,298	NA	NA
AY2019	653	856	1,032	1,237	1,235	NA	NA	NA

AY2020	639	863	1,294	1,249	NA	NA	NA	NA
AY2021	673	952	1,199	NA	NA	NA	NA	NA
AY2022	820	1,106	NA	NA	NA	NA	NA	NA
AY2023	799	NA	NA	NA	NA	NA	NA	NA

4 Methodology

4.1 Chain Ladder

The Chain Ladder method estimates age-to-age development factors using volume-weighted averages across accident years:

$$f_j = \frac{\sum_{i=1}^{n-j} C_{i,j+1}}{\sum_{i=1}^{n-j} C_{i,j}}$$

The cumulative development factor (CDF) from period j to ultimate is the product of all subsequent link ratios:

$$F_j = f_j \times f_{j+1} \times \cdots \times f_{n-1}$$

The projected ultimate for accident year i with latest observed development at period j^* is:

$$\hat{U}_i = C_{i,j^*} \times F_{j^*}$$

IBNR is then:

$$\widehat{IBNR}_i = \hat{U}_i - C_{i,j^*}$$

4.1.1 Mack Standard Error

Mack (1993) provides a distribution-free estimate of the reserve standard error, decomposing total uncertainty into process variance and parameter variance. The Mack standard error provides a measure of reserve uncertainty without requiring an assumed claims distribution — a key advantage over parametric alternatives.

4.2 Bornhuetter-Ferguson

The BF method estimates IBNR as:

$$\widehat{IBNR}_{BF,i} = \mu_i \times \left(1 - \frac{1}{F_{j^*}}\right)$$

where $\mu_i = P_i \times ELR^{AP}$ is the a priori ultimate (premium multiplied by the expected loss ratio), and $\left(1 - \frac{1}{F_{j^*}}\right)$ is the proportion of ultimate losses not yet emerged at the latest observed period.

The BF ultimate is:

$$\hat{U}_{BF,i} = C_{i,j^*} + \widehat{IBNR}_{BF,i}$$

4.2.1 A Priori ELR Selection

Table 6: A Priori ELR Selection and Sources

Line	A Priori ELR	Rationale
Motor	77%	IRDAI AR 2022-23: industry net incurred loss ratio for Motor ~76-78%
Engineering	70%	IRDAI Eng segment normalised for CAT years; non-CAT ELR ~68-72%
Treaty	73%	GIC Re AR 2022-23: net combined ratio ~98%, expense ratio ~25%

For Engineering, the a priori ELR of 70% deliberately excludes CAT year experience. This is the central purpose of BF in this context: providing a stable prior that is not contaminated by event-driven anomalies in the triangle.

4.3 Method Selection Framework

The preferred reserving method for each accident year is determined by the following rules, applied in order:

1. **CAT-contaminated AY** (Engineering AY2018, AY2019): BF preferred regardless of maturity. CL link ratios are distorted upward.
2. **Immature AY** (< 60% emerged): BF preferred. Insufficient emergence for CL to be credible.
3. **Mature AY** (90% emerged): CL preferred. Sufficient data.
4. **Intermediate AY** (60-90% emerged): Credibility blend — $w \times IBNR_{BF} + (1-w) \times IBNR_{CL}$ where $w = (1 - \% \text{ emerged})$.

5 Results

5.1 Development Factor Diagnostics

Before examining reserve estimates, it is important to assess whether the Chain Ladder assumption of stable development factors holds. The following charts show individual accident year link ratios against the volume-weighted average used by CL.

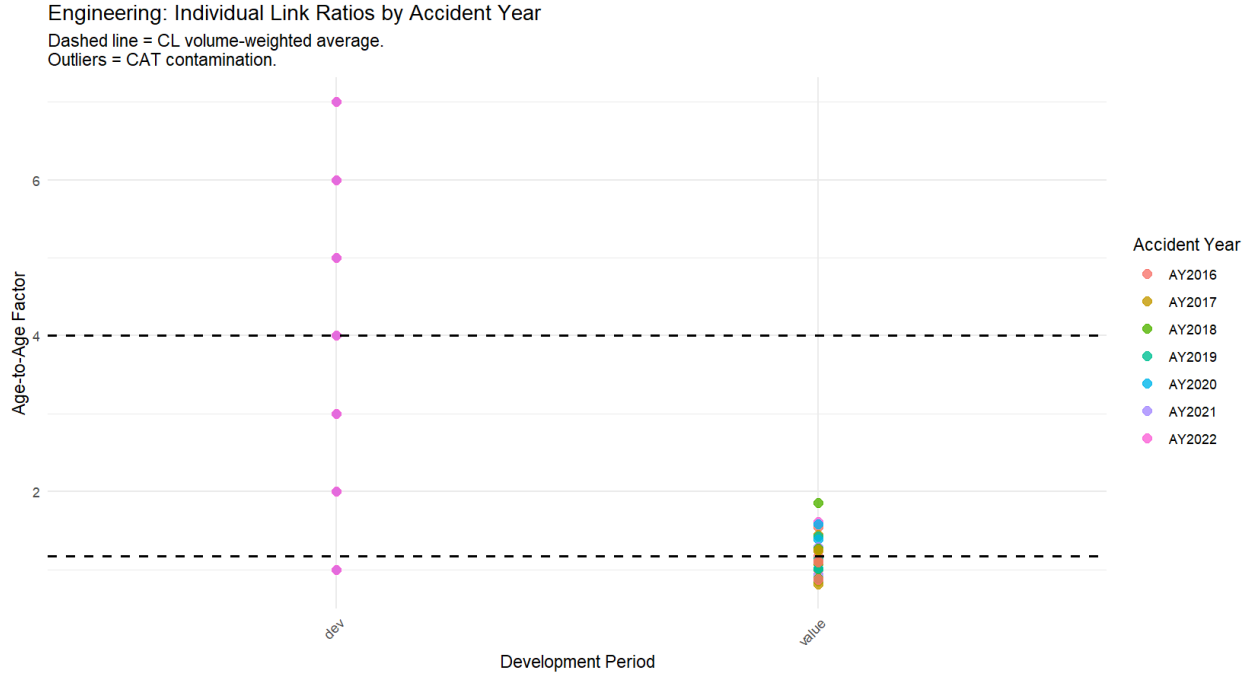


Figure 1: Engineering: Individual Link Ratios by Accident Year. Elevated AY2018 and AY2019 ratios (orange/red points above dashed line) reflect CAT contamination from Kerala floods and Cyclone Fani.

The Engineering chart is the critical diagnostic. AY2018 and AY2019 link ratios at Dev1→Dev2 are materially elevated above the CL average (dashed line). CL applies this inflated average to all accident years, including non-CAT future years — a systematic overstatement of Engineering reserves.

5.2 Chain Ladder Results

Table 7: Chain Ladder Results by Line and Accident Year (INR Crores)

Line	AY	Premium	Latest Obs	CL Ultimate	CL IBNR	Ult LR%
Motor						
Motor	AY2016	8,500	6,510.4	6,510.4	0.0	76.6
Motor	AY2017	9,520	7,202.8	7,137.9	-64.9	75.0
Motor	AY2018	10,660	8,228.5	8,445.9	217.4	79.2
Motor	AY2019	11,940	8,179.8	8,665.5	485.6	72.6

Motor	AY2020	13,370	9,234.4	10,175.7	941.3	76.1
Motor	AY2021	14,970	9,468.5	11,577.3	2,108.9	77.3
Motor	AY2022	16,770	9,201.3	13,326.6	4,125.4	79.5
Motor	AY2023	18,780	6,398.5	14,093.4	7,694.9	75.0
Engineering						
Engineering	AY2016	420	225.0	225.0	0.0	53.6
Engineering	AY2017	460	386.4	339.9	-46.5	73.9
Engineering	AY2018	510	491.0	510.3	19.4	100.1
Engineering	AY2019	555	468.3	454.6	-13.6	81.9
Engineering	AY2020	600	565.9	551.4	-14.5	91.9
Engineering	AY2021	650	370.1	440.0	69.9	67.7
Engineering	AY2022	710	374.0	448.6	74.6	63.2
Engineering	AY2023	775	323.3	580.6	257.3	74.9
Treaty						
Treaty	AY2016	1,200	819.2	819.2	0.0	68.3
Treaty	AY2017	1,320	904.4	948.3	43.9	71.8
Treaty	AY2018	1,450	1,297.6	1,191.1	-106.5	82.1
Treaty	AY2019	1,595	1,234.6	1,197.3	-37.3	75.1
Treaty	AY2020	1,755	1,249.0	1,262.2	13.1	71.9
Treaty	AY2021	1,930	1,198.5	1,298.4	99.8	67.3
Treaty	AY2022	2,120	1,106.4	1,488.2	381.8	70.2
Treaty	AY2023	2,335	798.8	1,535.8	737.0	65.8

5.3 BF vs Chain Ladder Comparison

Table 8: BF vs Chain Ladder IBNR Comparison (INR Crores)

Line	AY	BF IBNR	CL IBNR	Difference	Diff %
Motor					
Motor	AY2016	0.0	0.0	0.0	NaN
Motor	AY2017	-66.6	-64.9	-1.7	2.7
Motor	AY2018	211.2	217.4	-6.2	-2.8
Motor	AY2019	515.3	485.6	29.7	6.1
Motor	AY2020	952.3	941.3	11.0	1.2
Motor	AY2021	2099.7	2108.9	-9.2	-0.4
Motor	AY2022	3997.3	4125.4	-128.1	-3.1
Motor	AY2023	7895.4	7694.9	200.5	2.6
Engineering					
Engineering	AY2016	0.0	0.0	0.0	NaN
Engineering	AY2017	-44.1	-46.5	2.4	-5.2
Engineering	AY2018	13.5	19.4	-5.9	-30.3
Engineering	AY2019	-11.7	-13.6	1.9	-14.3
Engineering	AY2020	-11.0	-14.5	3.5	-24.0
Engineering	AY2021	72.3	69.9	2.4	3.5
Engineering	AY2022	82.7	74.6	8.1	10.8
Engineering	AY2023	240.4	257.3	-16.9	-6.6
Treaty					
Treaty	AY2016	0.0	0.0	0.0	NaN
Treaty	AY2017	44.6	43.9	0.7	1.6
Treaty	AY2018	-94.7	-106.5	11.8	-11.1
Treaty	AY2019	-36.3	-37.3	1.0	-2.8
Treaty	AY2020	13.3	13.1	0.2	1.4
Treaty	AY2021	108.3	99.8	8.5	8.5
Treaty	AY2022	397.1	381.8	15.3	4.0

Treaty	AY2023	818.0	737.0	81.0	11.0
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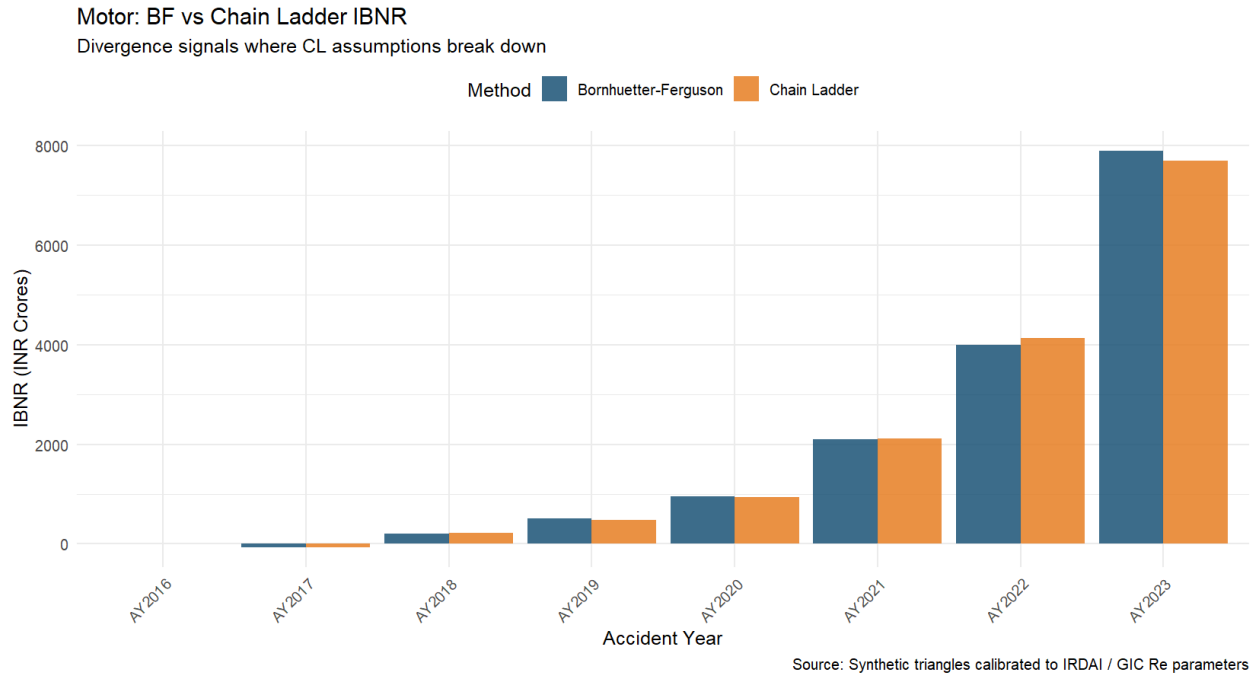


Figure 2: Motor: BF vs Chain Ladder IBNR. Methods converge — high volume yields stable link ratios, giving CL high credibility.

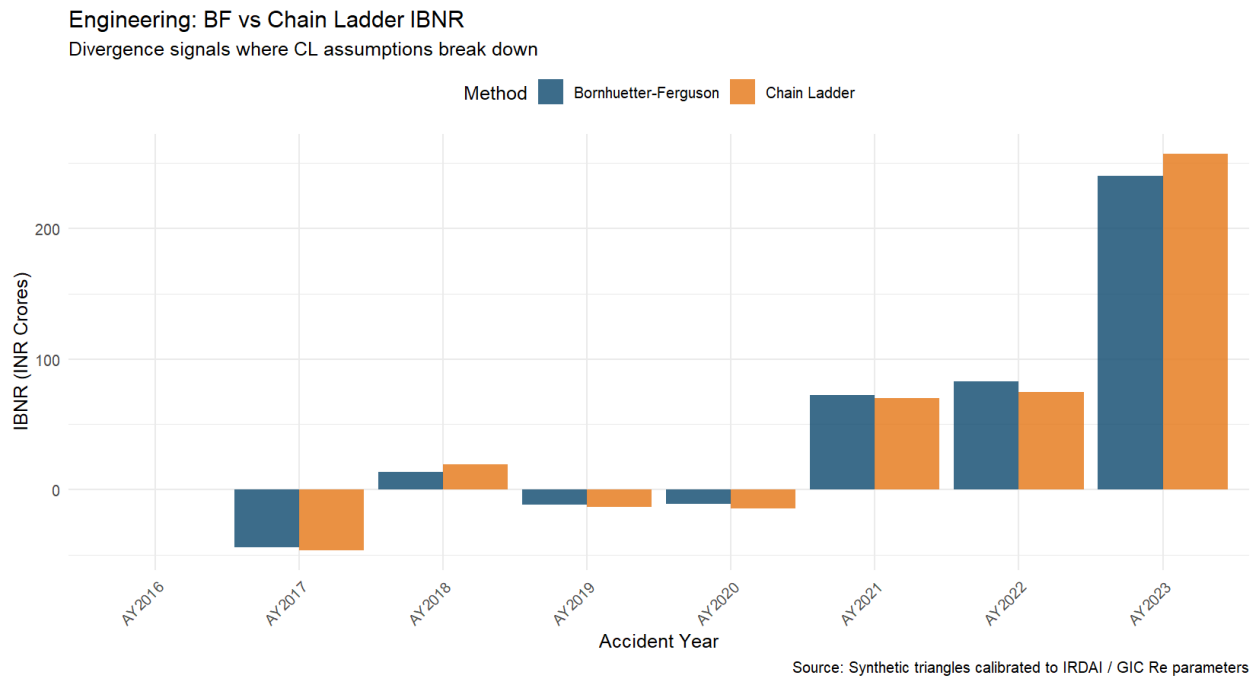


Figure 3: Engineering: BF vs Chain Ladder IBNR. Divergence in recent AYs reflects CAT contamination. BF is the preferred method.

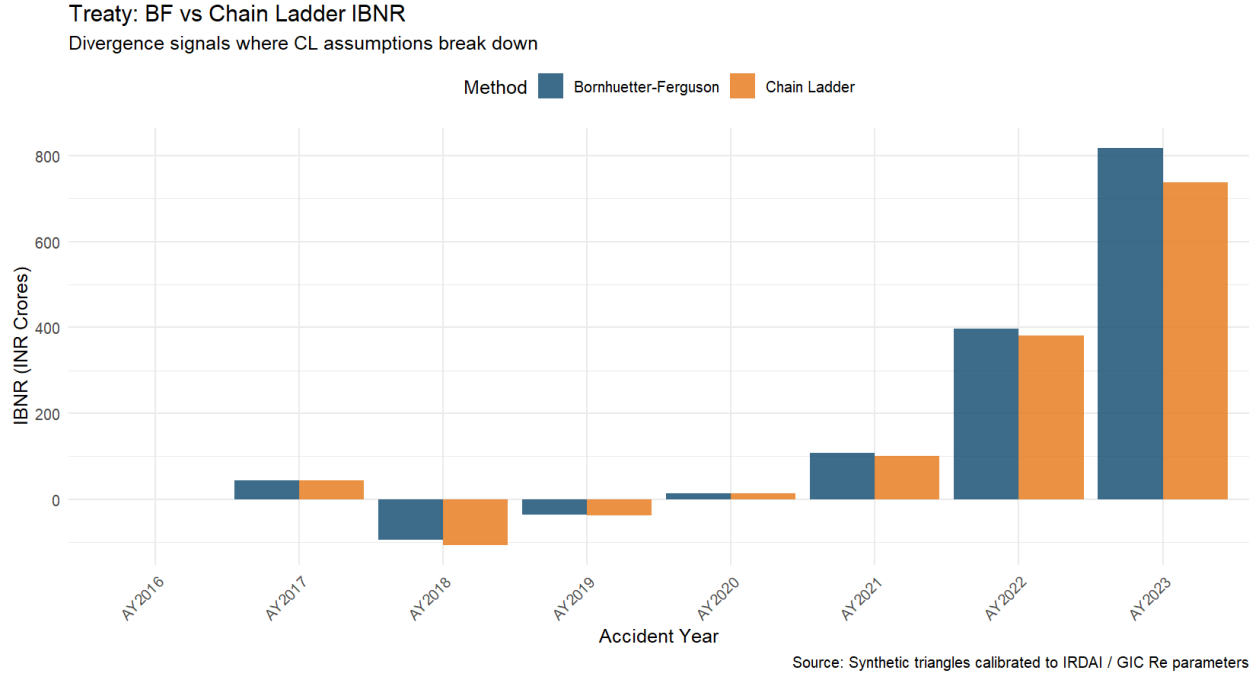


Figure 4: Treaty: BF vs Chain Ladder IBNR. BF produces higher estimates for immature AYs, reflecting greater reliance on the a priori ELR.

5.4 Portfolio Summary

Table 9: Portfolio Summary: Total IBNR by Line (INR Crores)

Line	Total Premium	Total Latest Obs	Total IBNR (CL)	Total IBNR (BF)	CL vs BF (%)
Engineering	4,680	3,204	347	342	1.3
Motor	104,510	64,424	15,509	15,605	-0.6
Treaty	13,705	8,608	1,132	1,250	-9.5

6 Key Findings and Actuarial Commentary

6.1 Finding 1: Motor — Methods Converge

CL and BF produce closely aligned IBNR estimates for Motor across all accident years. The high volume of Motor business yields stable, credible development factors. The minor divergence observed in AY2022 and AY2023 reflects BF giving more weight to the a priori in immature years — appropriate given limited emergence at those dates.

The Motor development pattern is materially slower than western benchmarks, consistent with MACT judicial delays. A practitioner applying western Motor TP development factors to an Indian book would understate the tail and underestimate IBNR for recent accident years.

6.2 Finding 2: Engineering — CAT Contamination Requires BF

This is the central actuarial finding of this report. The Kerala floods (2018) and Cyclone Fani (2019) generated elevated loss development in their respective accident years. CL incorporates these inflated link ratios into the volume-weighted average, which is then applied to all accident years — including future years where no comparable CAT is assumed.

The practical consequence: a practitioner using pure CL for Engineering without CAT exclusion will systematically overstate reserves for non-CAT years. BF, anchored to a normalised 70% a priori ELR, avoids this distortion. Engineering is the clearest example in this report of why method selection — not mechanical application of a single technique — is the core actuarial skill in reserving.

6.3 Finding 3: Treaty — Immature Year Stability

BF produces higher IBNR than CL for the two most recent treaty accident years (AY2022, AY2023). With only 25% and 48% of ultimate emerged respectively, CL has insufficient data to project credibly. BF’s reliance on the 73% a priori ELR for the unreported portion provides the stability that CL cannot.

This finding is consistent with standard actuarial practice: BF is generally preferred for accident years where less than 50-60% of ultimate has emerged.

7 Limitations and Extensions

7.1 Limitations of This Analysis

Synthetic data: All triangles are generated from calibrated parameters rather than actual insurer or reinsurer data. Results should be interpreted as illustrative of methodology, not as estimates of any specific entity’s reserves.

Tail factor: A tail factor of 1.000 is assumed for all lines, implying all development is complete by period 8. For Motor TP in India, given MACT delays, a tail factor greater than 1.0 may be warranted and should be investigated with actual data.

No calendar year effects: The Mack model assumes accident years are independent. In practice, Indian Motor TP reserves may be affected by calendar year effects (e.g., changes in MACT award levels, inflation in court settlements). A diagonal effects test (Thomas, 2006) would be appropriate for production use.

CAT year exclusion: This analysis flags AY2018 and AY2019 as CAT-contaminated and uses BF. In practice, an actuary would also consider excluding these years from the link ratio calculation entirely, or applying a weighted average that downweights event years.

7.2 Suggested Extensions

- **Generalised Linear Model (GLM) reserving** via `glmReserve()` in `ChainLadder` — provides a parametric alternative with explicit distributional assumptions
 - **Munich Chain Ladder** — uses both paid and incurred triangles simultaneously, appropriate where both are available
 - **CAT-adjusted link ratios** — explicitly exclude or downweight AY2018/2019 from the Engineering link ratio calculation and compare results
 - **Tail factor estimation** — fit a curve (e.g., inverse power, exponential) to the development pattern to extrapolate beyond period 8 for Motor TP
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8 References

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This report was produced using R (version 4.6.0) and Quarto. All analysis is reproducible from the accompanying GitHub repository.

Methodology questions or feedback welcome via LinkedIn.